Exercise Sheet 3
Date of discussion: 12. November 2026
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Question 1: Cryptographic Hash Functions

a) What is the difference between a cryptographic hash function and a cyclic redundancy check (CRC)? Why is a CRC unsuitable for detecting malicious modifications?

b) Explain why finding a collision in a cryptographic hash function is unavoidable in principle. Does this mean that cryptographic hash functions are insecure?

c) Consider a cryptographic hash function producing a 64-bit hash value.

   a) Approximately how many attempts are needed to find a pre-image of a given hash value?

   b) Approximately how many randomly selected messages are needed to find a collision using the birthday attack?

   c) How would both results change for a 128-bit hash function?

Question 2: Symmetric Cryptography

a) Explain the basic principle of symmetric encryption. What is the relationship between the encryption key and the decryption key? Express the encryption and decryption process using the notation from the lecture.

b) Alice and Bob communicate using symmetric encryption. An attacker can intercept all messages exchanged over the network but does not know their shared secret key.

   a) Which information should the attacker be unable to obtain from the ciphertext?

   b) Can the attacker prevent Bob from receiving Alice's messages?

   c) What happens if the attacker obtains the secret key?

c) Alice, Bob, and Charlie use symmetric cryptography. Alice shares a secret key K_{AB} with Bob, and Bob shares another secret key K_{BC} with Charlie. Alice wants to send a confidential message m to Charlie through Bob.

a) Describe the communication using the encryption and decryption notation from the lecture.

b) Can Bob read the message?

c) How could Alice communicate confidentially with Charlie without revealing the plaintext to Bob?